

# Lecture 11: Krylov subspace, Generalized minimal residual method



School of Mathematical Sciences, Xiamen University

## 1. Krylov subspace

- Given nonzero  $\mathbf{A} \in \mathbb{C}^{m \times m}$  and nonzero  $\mathbf{r} \in \mathbb{C}^m$ , the  $j$ th Krylov subspace generated by  $\mathbf{A}$  and  $\mathbf{r}$  is defined by

$$\mathcal{K}_j(\mathbf{A}, \mathbf{r}) := \text{span}\{\mathbf{r}, \mathbf{A}\mathbf{r}, \mathbf{A}^2\mathbf{r}, \dots, \mathbf{A}^{j-1}\mathbf{r}\}.$$

Obviously,  $\mathcal{K}_j(\mathbf{A}, \mathbf{r}) \subseteq \mathcal{K}_{j+1}(\mathbf{A}, \mathbf{r})$  and  $\dim \mathcal{K}_j(\mathbf{A}, \mathbf{r}) \leq j$ .

### Proposition 1

Let  $\mathbb{P}_j$  denote the set of polynomials of degree  $\leq j$ . Then

$$\mathcal{K}_j(\mathbf{A}, \mathbf{r}) = \{p(\mathbf{A})\mathbf{r} \mid p \in \mathbb{P}_{j-1}\}.$$

### Proposition 2

If the minimal polynomial of the matrix  $\mathbf{A}$  has degree  $n$ , then for any  $j > n$  and any nonzero  $\mathbf{r} \in \mathbb{C}^m$ , we have

$$\dim \mathcal{K}_j(\mathbf{A}, \mathbf{r}) \leq n.$$

## 1.1. Arnoldi process

---

**Algorithm:** Arnoldi process generating orthonormal basis

---

Given nonzero  $\mathbf{A} \in \mathbb{C}^{m \times m}$  and nonzero  $\mathbf{r} \in \mathbb{C}^m$

$$\mathbf{q}_1 = \mathbf{r} / \|\mathbf{r}\|_2$$

**for**  $j = 1, 2, 3, \dots$ ,

$$\mathbf{v} = \mathbf{A}\mathbf{q}_j$$

**for**  $i = 1$  **to**  $j$

$$h_{ij} = \langle \mathbf{v}, \mathbf{q}_i \rangle = \mathbf{q}_i^* \mathbf{v}$$

$$\mathbf{v} = \mathbf{v} - h_{ij} \mathbf{q}_i$$

**end**

$$h_{j+1,j} = \|\mathbf{v}\|_2$$

$$\mathbf{q}_{j+1} = \mathbf{v} / h_{j+1,j}$$

**end**

---

- At the end of step  $j$ , we obtain

$$\mathbf{v} = (\mathbf{I} - \mathbf{q}_j \mathbf{q}_j^*) \cdots (\mathbf{I} - \mathbf{q}_2 \mathbf{q}_2^*) (\mathbf{I} - \mathbf{q}_1 \mathbf{q}_1^*) \mathbf{A} \mathbf{q}_j.$$

- We call the Arnoldi process breaks down at step  $k$  if  $h_{k+1,k} = 0$ .

### Remark 3

*The Arnoldi process is the modified Gram–Schmidt orthogonalization applied to  $\{\mathbf{r}, \mathbf{A}\mathbf{q}_1, \mathbf{A}\mathbf{q}_2, \dots, \mathbf{A}\mathbf{q}_k\}$ . We have the Arnoldi relation*

$$\mathbf{A} [\mathbf{q}_1 \quad \cdots \quad \mathbf{q}_j] = [\mathbf{q}_1 \quad \cdots \quad \mathbf{q}_{j+1}] \begin{bmatrix} h_{11} & \cdots & h_{1j} \\ h_{21} & \ddots & \vdots \\ & \ddots & h_{jj} \\ & & & h_{j+1,j} \end{bmatrix}, \quad \forall 1 \leq j < k,$$

*that is  $\mathbf{A}\mathbf{Q}_j = \mathbf{Q}_{j+1}\tilde{\mathbf{H}}_j$ . ( $\tilde{\mathbf{H}}_j$  is upper Hessenberg.) Let*

$$\mathbf{H}_j := \begin{bmatrix} h_{11} & \cdots & h_{1j} \\ h_{21} & \ddots & \vdots \\ & \ddots & h_{jj} \end{bmatrix}, \quad \forall 1 \leq j \leq k.$$

*We have  $\mathbf{H}_j = \mathbf{Q}_j^* \mathbf{A} \mathbf{Q}_j$  for all  $1 \leq j \leq k$ , and  $\mathbf{A} \mathbf{Q}_k = \mathbf{Q}_k \mathbf{H}_k$ .*

## Theorem 4

*Suppose that the Arnoldi process breaks down at step  $k$ . Then the set  $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k\}$  is orthonormal and for  $j = 1, 2, \dots, k$ ,*

$$\text{span}\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_j\} = \mathcal{K}_j(\mathbf{A}, \mathbf{r}), \quad \dim \mathcal{K}_j(\mathbf{A}, \mathbf{r}) = j.$$

*Moreover, for  $j > k$ , we have  $\mathcal{K}_j(\mathbf{A}, \mathbf{r}) = \mathcal{K}_k(\mathbf{A}, \mathbf{r})$  and  $\dim \mathcal{K}_j(\mathbf{A}, \mathbf{r}) = k$ .*

## Corollary 5

*The matrices  $\{\mathbf{Q}_j\}_{j=1}^k$  generated by the Arnoldi process are Q-factors of reduced QR factorizations of the Krylov matrices,*

$$\mathbf{K}_j := [\mathbf{r} \quad \mathbf{A}\mathbf{r} \quad \dots \quad \mathbf{A}^{j-1}\mathbf{r}] = \mathbf{Q}_j \mathbf{R}_j, \quad j = 1, 2, \dots, k.$$

**Exercise:** Both  $\mathbf{K}_j$  and  $\mathbf{R}_j$  are not formed explicitly in the Arnoldi process. We have  $r_{11} = \|\mathbf{r}\|_2$ . How to obtain  $\mathbf{R}_j$  from  $\mathbf{H}_j$ ?

**Exercise:** Prove the implicit Q theorem via Krylov subspaces.

## 2. Generalized minimal residual method (GMRES)

- The principle of GMRES: Consider a nonsingular linear system

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{C}^{m \times m}, \quad \mathbf{b} \in \mathbb{C}^m.$$

For any initial guess  $\mathbf{x}_0$ , at step  $j$ , GMRES finds the  $j$ th approximate solution

$$\mathbf{x}_j = \operatorname{argmin}_{\mathbf{x} \in \mathbf{x}_0 + \mathcal{K}_j(\mathbf{A}, \mathbf{r}_0)} \|\mathbf{b} - \mathbf{Ax}\|_2,$$

where  $\mathbf{r}_0 := \mathbf{b} - \mathbf{Ax}_0$  and

$$\mathcal{K}_j(\mathbf{A}, \mathbf{r}_0) = \operatorname{span}\{\mathbf{r}_0, \mathbf{Ar}_0, \dots, \mathbf{A}^{j-1}\mathbf{r}_0\}.$$

For the residual  $\mathbf{r}_j := \mathbf{b} - \mathbf{Ax}_j$ , we have

$$\|\mathbf{r}_j\|_2 = \min_{\mathbf{x} \in \mathbf{x}_0 + \mathcal{K}_j(\mathbf{A}, \mathbf{r}_0)} \|\mathbf{b} - \mathbf{Ax}\|_2.$$

Assume that the Arnoldi process for the orthonormal basis of  $\mathcal{K}_j(\mathbf{A}, \mathbf{r}_0)$  breaks down at step  $k$ . For  $1 \leq j < k$ , we have

$$\begin{aligned}\|\mathbf{r}_j\|_2 &= \min_{\mathbf{x} \in \mathbf{x}_0 + \mathcal{K}_j(\mathbf{A}, \mathbf{r}_0)} \|\mathbf{b} - \mathbf{A}\mathbf{x}\|_2 = \min_{\mathbf{y} \in \mathbb{C}^j} \|\mathbf{r}_0 - \mathbf{A}\mathbf{Q}_j\mathbf{y}\|_2 \\ &= \min_{\mathbf{y} \in \mathbb{C}^j} \left\| \mathbf{r}_0 - \mathbf{Q}_{j+1}\tilde{\mathbf{H}}_j\mathbf{y} \right\|_2 \quad (\text{by Arnoldi relation}) \\ &= \min_{\mathbf{y} \in \mathbb{C}^j} \left\| \|\mathbf{r}_0\|_2 \mathbf{e}_1 - \tilde{\mathbf{H}}_j\mathbf{y} \right\|_2 > 0.\end{aligned}$$

For  $j = k$ , we have

$$\begin{aligned}\|\mathbf{r}_k\|_2 &= \min_{\mathbf{x} \in \mathbf{x}_0 + \mathcal{K}_k(\mathbf{A}, \mathbf{r}_0)} \|\mathbf{b} - \mathbf{A}\mathbf{x}\|_2 = \min_{\mathbf{y} \in \mathbb{C}^k} \|\mathbf{r}_0 - \mathbf{A}\mathbf{Q}_k\mathbf{y}\|_2 \\ &= \min_{\mathbf{y} \in \mathbb{C}^k} \|\mathbf{r}_0 - \mathbf{Q}_k\mathbf{H}_k\mathbf{y}\|_2 \quad (\text{by Arnoldi relation}) \\ &= \min_{\mathbf{y} \in \mathbb{C}^k} \left\| \|\mathbf{r}_0\|_2 \mathbf{e}_1 - \mathbf{H}_k\mathbf{y} \right\|_2 = 0.\end{aligned}$$

Once  $\mathbf{y}_j$  is found, set  $\mathbf{x}_j = \mathbf{x}_0 + \mathbf{Q}_j\mathbf{y}_j$ .

- The least squares problem about  $\mathbf{y}$  can be solved inexpensively with Givens rotations, exploiting the upper Hessenberg structure of  $\tilde{\mathbf{H}}_j$ , costing just  $\mathcal{O}(j^2)$  instead of  $\mathcal{O}(j^3)$ .

## 2.1. Convergence of GMRES

### Theorem 6

*Assume that the Arnoldi process for the orthonormal basis of  $\mathcal{K}_j(\mathbf{A}, \mathbf{r}_0)$  breaks down at step  $k$ .*

- (1) *For  $1 \leq j < k$ , the residual  $\mathbf{r}_j$  satisfies  $(\mathbf{A}\mathbf{Q}_j)^*\mathbf{r}_j = \mathbf{0}$ , i.e.,*

$$\mathbf{r}_j \perp \mathbf{A}\mathcal{K}_j.$$

- (2) *For  $0 \leq j \leq k$ , the residual  $\mathbf{r}_j$  satisfies*

$$\|\mathbf{r}_0\|_2 \geq \|\mathbf{r}_1\|_2 \geq \cdots \geq \|\mathbf{r}_{k-1}\|_2 > \|\mathbf{r}_k\|_2 = 0.$$

*That is to say GMRES converges monotonically and finds the exact solution at step  $k$ .*

## Theorem 7

Suppose  $\mathbf{A}$  is diagonalizable, i.e.,  $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$  for some nonsingular matrix  $\mathbf{V}$  and diagonal matrix  $\mathbf{\Lambda}$ . At step  $j$  of the GMRES iteration, the residual  $\mathbf{r}_j$  satisfies

$$\frac{\|\mathbf{r}_j\|_2}{\|\mathbf{r}_0\|_2} \leq \min_{p \in \mathbb{P}_j, p(0)=1} \|p(\mathbf{A})\|_2 \leq \kappa(\mathbf{V}) \min_{p \in \mathbb{P}_j, p(0)=1} \max_{\lambda \in \Lambda(\mathbf{A})} |p(\lambda)|,$$

where  $\Lambda(\mathbf{A})$  is the set of eigenvalues of  $\mathbf{A}$ , and  $\kappa(\mathbf{V}) = \|\mathbf{V}\|_2 \|\mathbf{V}^{-1}\|_2$ .

- Y. Saad and M.H. Schultz

GMRES: A generalized minimal residual algorithm for solving non-symmetric linear systems

SIAM J. Sci. Stat. Comput., 7: 856–869, 1986.

- Y. Saad

A flexible inner-outer preconditioned GMRES algorithm

SIAM J. Sci. Comput., 14: 461–469, 1993.

- **Exercise:** Assume the Arnoldi process for  $\{\mathbf{A}, \mathbf{r}_0\}$  breaks down at step  $k > 1$ . For  $1 \leq j < k$ , we have the Arnoldi relation

$$\mathbf{A}\mathbf{Q}_j = \mathbf{Q}_{j+1}\tilde{\mathbf{H}}_j.$$

For  $1 \leq j < k$ , prove the following:

- (a) The  $j$ th residual  $\mathbf{r}_j$  of GMRES can be *uniquely* expressed as

$$\mathbf{r}_j = p_j(\mathbf{A})\mathbf{r}_0, \quad \deg(p_j) \leq j, \quad p_j(0) = 1.$$

- (b) Let  $\mathbf{H}_j = \mathbf{Q}_j^* \mathbf{A} \mathbf{Q}_j$ . The unique polynomial  $p_j$  in (a) is given by

$$p_j(z) = \prod_{i=1}^j \left(1 - \theta_i^{(j)} z\right),$$

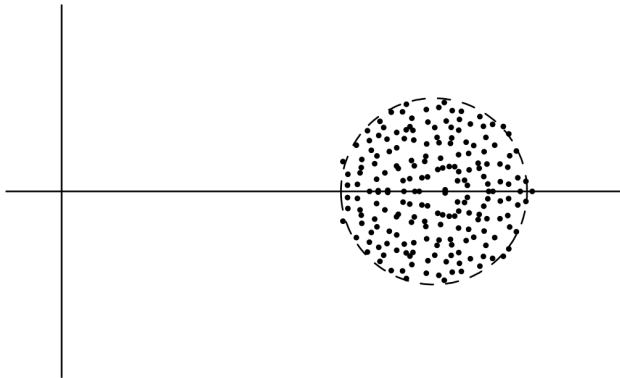
where  $\theta_i^{(j)}$ ,  $i = 1, 2, \dots, j$ , are the eigenvalues of  $(\tilde{\mathbf{H}}_j^* \tilde{\mathbf{H}}_j)^{-1} \mathbf{H}_j^*$ .

## 2.2. Numerical examples

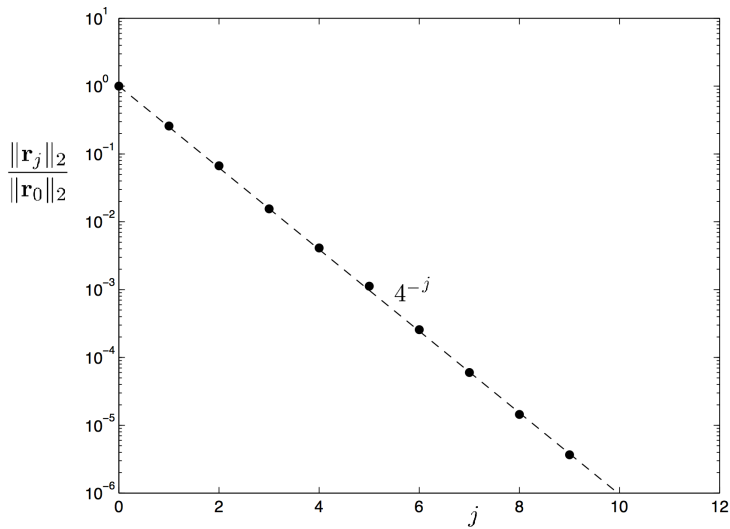
- Example 1:  $\mathbf{A}$ ,  $200 \times 200$  entries from real normal distribution of mean 2 and standard deviation  $0.5/\sqrt{200}$

$m = 200$ ;  $\mathbf{A} = 2*\text{eye}(m)+0.5*\text{randn}(m)/\text{sqrt}(m)$ ;

$\mathbf{Ax} = \mathbf{b}$ ,  $\mathbf{x}_0 = \mathbf{0}$ ,  $\mathbf{b} = [1 \ 1 \ \dots \ 1]^\top$



## Convergence history of Example 1

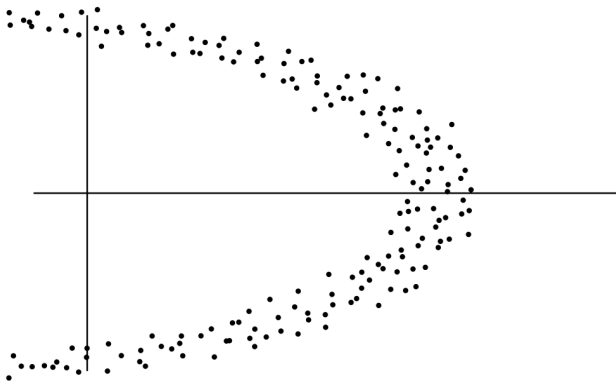


• Example 2:

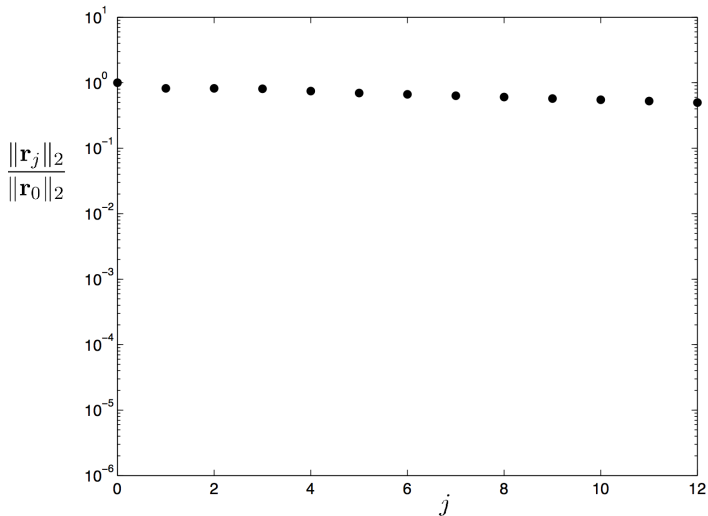
$m = 200$ ;  $B = 2 \cdot \text{eye}(m) + 0.5 \cdot \text{randn}(m) / \text{sqrt}(m)$ ;

$A = B + D$ ,  $D$  is the diagonal matrix with complex entries

$$d_i = (-2 + 2 \sin \theta_i) + i \cos \theta_i, \quad \theta_i = \frac{(i-1)\pi}{m-1}, \quad 1 \leq i \leq m.$$



## Convergence history of Example 2



## 2.3. Preconditioning (see Lecture 40 of NLA)

- To improve the convergence of Krylov subspace methods, it is important to have a preconditioner, denoted by  $\mathbf{M}$ .
- Left preconditioning, i.e.,

$$\mathbf{M}^{-1}\mathbf{A}\mathbf{x} = \mathbf{M}^{-1}\mathbf{b}.$$

- Right preconditioning is often used, i.e.,

$$\mathbf{A}\mathbf{M}^{-1}\mathbf{z} = \mathbf{b}, \quad \mathbf{x} = \mathbf{M}^{-1}\mathbf{z},$$

because it produces the same residual as that of the original system in exact precision arithmetic.

- The preconditioned matrix  $\mathbf{M}^{-1}\mathbf{A}$  or  $\mathbf{A}\mathbf{M}^{-1}$  should have eigenvalues clustering behavior.
- Only the action of applying  $\mathbf{M}^{-1}$  to a given vector is computed in GMRES. So we never explicitly form  $\mathbf{M}^{-1}$ . We only require that the action  $\mathbf{M}^{-1}\mathbf{z}$  must be cheap.

- How to find a good preconditioner? **It's problem dependent.**

**Example.** Let  $\mathcal{A} = \begin{bmatrix} \mathbf{A} & \mathbf{B}^* \\ \mathbf{C} & \mathbf{0} \end{bmatrix}$  and  $\mathcal{M} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}\mathbf{A}^{-1}\mathbf{B}^* \end{bmatrix}$ , where  $\mathbf{A} \in \mathbb{C}^{m \times m}$  is invertible, and  $\mathbf{B}, \mathbf{C} \in \mathbb{C}^{n \times m}$  with  $m \geq n$ . Assume that  $-\mathbf{C}\mathbf{A}^{-1}\mathbf{B}^*$  is invertible.

The preconditioned matrix  $\mathcal{M}^{-1}\mathcal{A}$  is diagonalizable and has at most three distinct eigenvalues 1,  $(1 + \sqrt{5})/2$ , and  $(1 - \sqrt{5})/2$ .

## 2.4. Restarted GMRES

- For larger values of  $j$ , the cost of GMRES in operations and storage may be prohibitive. In such circumstances a method called  $l$ -step restarted GMRES or GMRES( $l$ ) is often employed.
- GMRES( $l$ ): After  $l$  steps, the GMRES iteration is started anew with the current vector  $\mathbf{x}_l$  as an initial guess.
- Note that GMRES( $l$ ) can be expected fail to converge, whereas GMRES always succeeds for exact arithmetic. (**Embree's paper**)
- **GMRES-IR**, **GMRES-DR**, **FGMRES-DR**, etc.